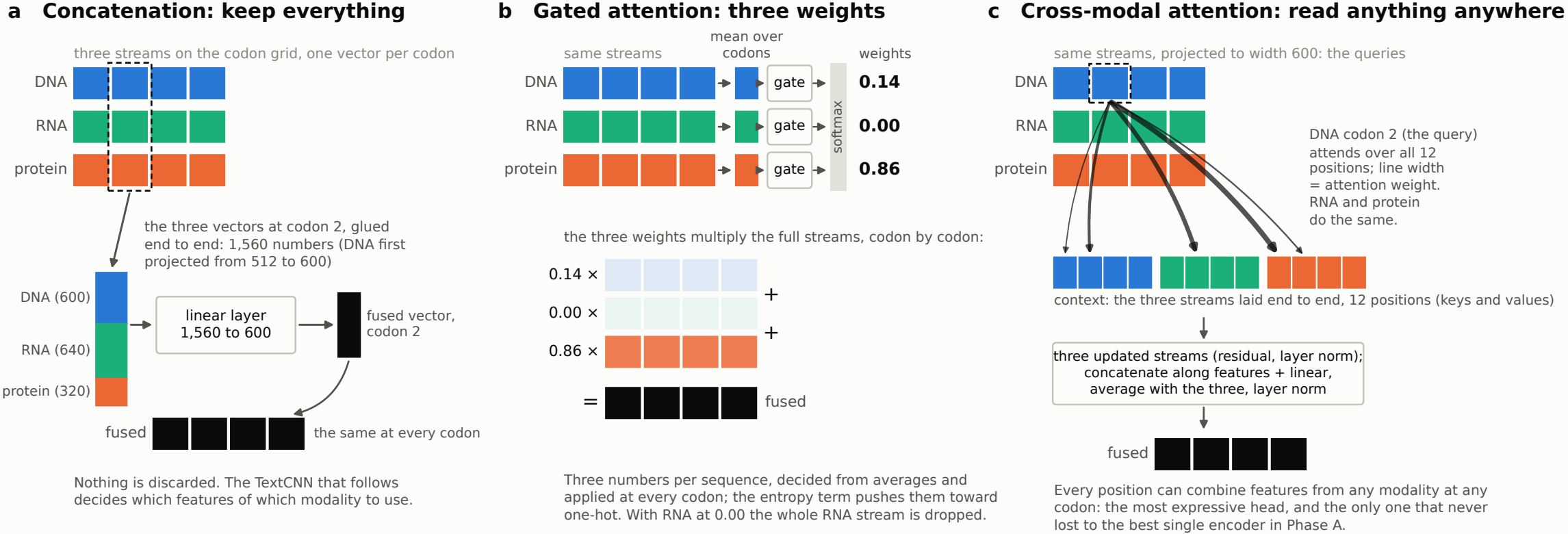
a Concatenation: keep everything
three streams on the codon grid, one vector per codon
DNA
RNA
protein
the three vectors at codon 2, glued end to end: 1,560 numbers (DNA first projected from 512 to 600)
DNA (600)
RNA (640)
protein (320)
linear layer 1,560 to 600
fused vector, codon 2
fused
the same at every codon
Nothing is discarded. The TextCNN that follows decides which features of which modality to use.
b Gated attention: three weights
same streams
mean over codons
weights
DNA
RNA
protein
gate
gate
gate
softmax
0.14
0.00
0.86
the three weights multiply the full streams, codon by codon:
0.14 ×
0.00 ×
0.86 ×
+
+
=
fused
Three numbers per sequence, decided from averages and applied at every codon; the entropy term pushes them toward one-hot. With RNA at 0.00 the whole RNA stream is dropped.
c Cross-modal attention: read anything anywhere
same streams, projected to width 600: the queries
DNA
RNA
protein
DNA codon 2 (the query) attends over all 12 positions; line width = attention weight. RNA and protein do the same.
context: the three streams laid end to end, 12 positions (keys and values)
three updated streams (residual, layer norm); concatenate along features + linear, average with the three, layer norm
fused
Every position can combine features from any modality at any codon: the most expressive head, and the only one that never lost to the best single encoder in Phase A.